

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S6 (R, S) / S6 (PT) (R) Examination June 2023 (2019 Scheme)

Course Code: CST306**Course Name: ALGORITHM ANALYSIS AND DESIGN**

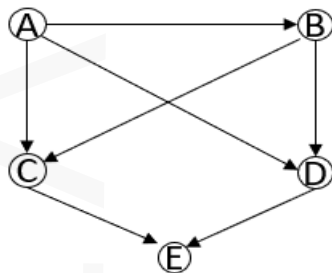
Max. Marks: 100

Duration: 3 Hours

PART A*Answer all questions, each carries 3 marks.*

Marks

- 1 Show that for any real constants a and b, where $b > 0$, $(n + a)^b = O(n^b)$ (3)
- 2 Solve the following recurrence equations using Master theorem.
 a. $T(n) = 3T(n/2) + n^2$ (3)
 b. $T(n) = 2T(n/2) + n \log n$
- 3 Define AVL tree. Explain the rotations performed for insertion in AVL tree. (3)
- 4 Find the different topological ordering of the given graph. (3)



- 5 Write the control abstraction of divide and conquer strategy. (3)
- 6 Compare Strassen's matrix multiplication with ordinary matrix multiplication. (3)
- 7 Differentiate backtracking technique from branch and bound technique. (3)
- 8 What is Principle of Optimality? (3)
- 9 Differentiate P and NP problems. Give one example to each. (3)
- 10 Define graph coloring problem. (3)

PART B*Answer one full question from each module, each carries 14 marks.***Module I**

- 11 a) Define Big Oh, Big Omega and Theta notations and illustrate them graphically. (7)
- b) Find the time complexity of following code segment (7)


```

      (i) for (int i = 1; i <= n; i *= c) {
          // some O(1) expressions
      }
      
```

```

    for (int i = n; i > 0; i /= c) {
        // some O(1) expressions
    }
(ii) for (int i = 1; i <= n; i += c) {
    // some O(1) expressions
}
for (int i = n; i > 0; i -= c) {
    // some O(1) expressions
}

```

OR

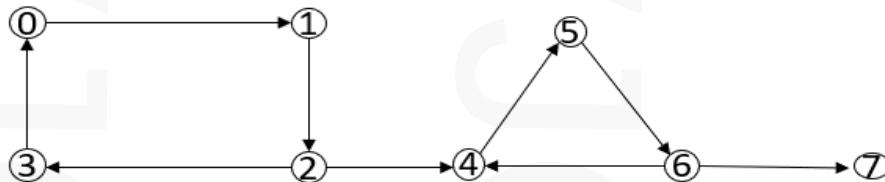
- 12 a) Find the best case, worst case and average case time complexity of binary search. (7)
- b) Find the time complexity of following function using recursion tree method. (7)
- (i) $T(n) = 2T(n/2) + n^2$
- (ii) $T(n) = T(n/3) + T(2n/3) + n$

Module II

- 13 a) Construct AVL tree by inserting following elements appeared in the order. (7)
21, 26, 30, 9, 4, 14, 28, 18, 15
- b) Explain union and find algorithms in disjoint datasets. (7)

OR

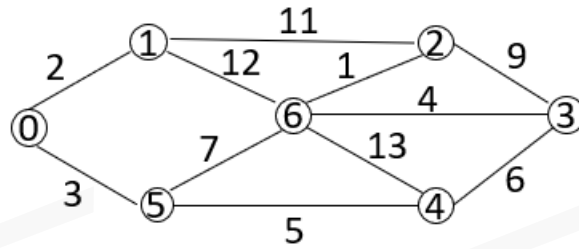
- 14 a) Write DFS algorithm for graph traversal. Also derive its time complexity. (7)
- b) Find the strongly connected components of the given directed graph. (7)

**Module III**

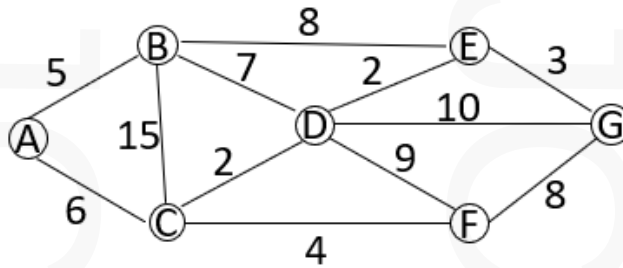
- 15 a) Explain 2- way merge sort algorithm with an example and derive its time complexity. (7)
- b) Find the optimal solution for the following Fractional Knapsack problem. (7)
Given the number of items(n) = 7, capacity of sack(m) = 15,
 $W = \{1, 3, 5, 4, 1, 3, 2\}$ and $P = \{10, 15, 7, 8, 9, 4\}$

OR

- 16 a) Apply Kruskal's algorithm for finding minimum cost spanning tree. (7)



- b) Apply Dijkstra's algorithm for finding the shortest path from vertex A to all other vertices. (7)

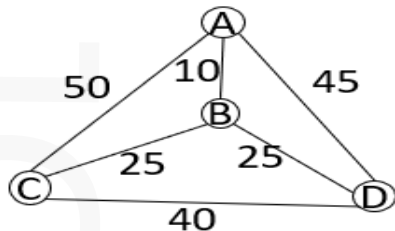


Module IV

- 17 a) Find the optimal parenthesis of matrix chain product whose sequence of dimensions is $5 \times 4, 4 \times 6, 6 \times 2, 2 \times 7$ (8)
 b) Explain 4 queen problem. Draw the state space tree for 4 queen problem. (6)

OR

- 18 a) Define TSP problem. Apply branch and bound algorithm for solving TSP. (9)



- b) Write Floyd Warshall's algorithm for finding all pairs shortest path algorithm. (5)

Module V

- 19 a) Explain the first fit-decreasing strategy of bin packing algorithm. (7)
 b) Prove that Clique Decision problem is NP-complete. (7)

OR

- 20 a) Differentiate Las Vegas and Monte Carlo algorithms (7)
 b) Explain randomized quick sort with the help of suitable examples. (7)
